



NC STATE
UNIVERSITY

NRP 
Nuclear Reactor Program

NE235 Nuclear Reactor Operations Module #9

Reactivity Coefficients

Dept. of Nuclear Engineering
North Carolina State University

NE235 Nuclear Reactor Operations Training **NRP**

Module #9

Module #9 – Reactivity Coefficients:

- A. Moderator Temperature Coefficient
- B. Void Coefficient
- C. Doppler Broadening (Fuel Temperature) Coefficient
- D. Fission Product Poisons

Laboratory Session this week:

- No lab this week!

Homework #9 – Due 11/08

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Module #9

Net Reactivity is comprised of the following reactivity parameters which we will cover in the next few lectures:

$$\rho_{\text{net}} = \rho_{\text{excess}} + \rho_{\text{rods}} + \rho_{\text{Doppler}} + \rho_{\text{mod temp}} + \rho_{\text{voids}} + \rho_{\text{poisons}} + \rho_{\text{experiments}} + \rho_{\text{source}}$$

For this lecture we will focus on:

- $\rho_{\text{mod temp}}$: reactivity effects due to changes in moderator temperature
- ρ_{voids} : reactivity effects due to moderator displacement by voids
- ρ_{Doppler} : reactivity effects due to change in fuel temperature
- ρ_{poisons} : reactivity effects due to addition of fission products that have large neutron absorption cross sections

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Module #9

Reactivity Coefficients:

- A. Moderator Temperature Coefficient : $\alpha_{T_{Mod}} (\text{pcm}/^{\circ}\text{F}_{mod})$
- B. Void Coefficient : $\alpha_{voids} (\text{pcm}/\text{cc}_{void})$
- C. Fuel Temp Coefficient (Doppler) : $\alpha_{Fuel} (\text{pcm}/^{\circ}\text{F}_{fuel})$

How do these α coefficients vary as a function of operating conditions of the reactor over time?

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A. Moderator Temperature Coefficient : $\alpha_{T \text{ mod}}$

Function of 3 sub-coefficients relating to the 6 factor formula:

$$\alpha_{T \text{ mod}} = \underbrace{\alpha_T(f)}_{\#1} + \underbrace{\alpha_T(p)}_{\#2} + \underbrace{\alpha_T(\mathcal{L})}_{\#3} = \frac{\Delta\rho(\text{pcm})}{^\circ\text{F}_{\text{mod}}}$$

#1: Thermal Utilization (f)

#2: Resonance Escape Probability (p)

#3: Non-Leakage Probabilities ($\mathcal{L}$)

Moderator Temperature Coefficient

#1: Thermal Utilization Factor (f):

$$f \equiv \frac{\Sigma_{a \text{ fuel}}}{\Sigma_{a \text{ core}}} = \frac{\Sigma_{a \text{ fuel}}}{\Sigma_{a \text{ fuel}} + \Sigma_{a \text{ coolant/mod}} + \Sigma_{a \text{ core structures}}}$$

where $\Sigma_a = N\sigma_a$

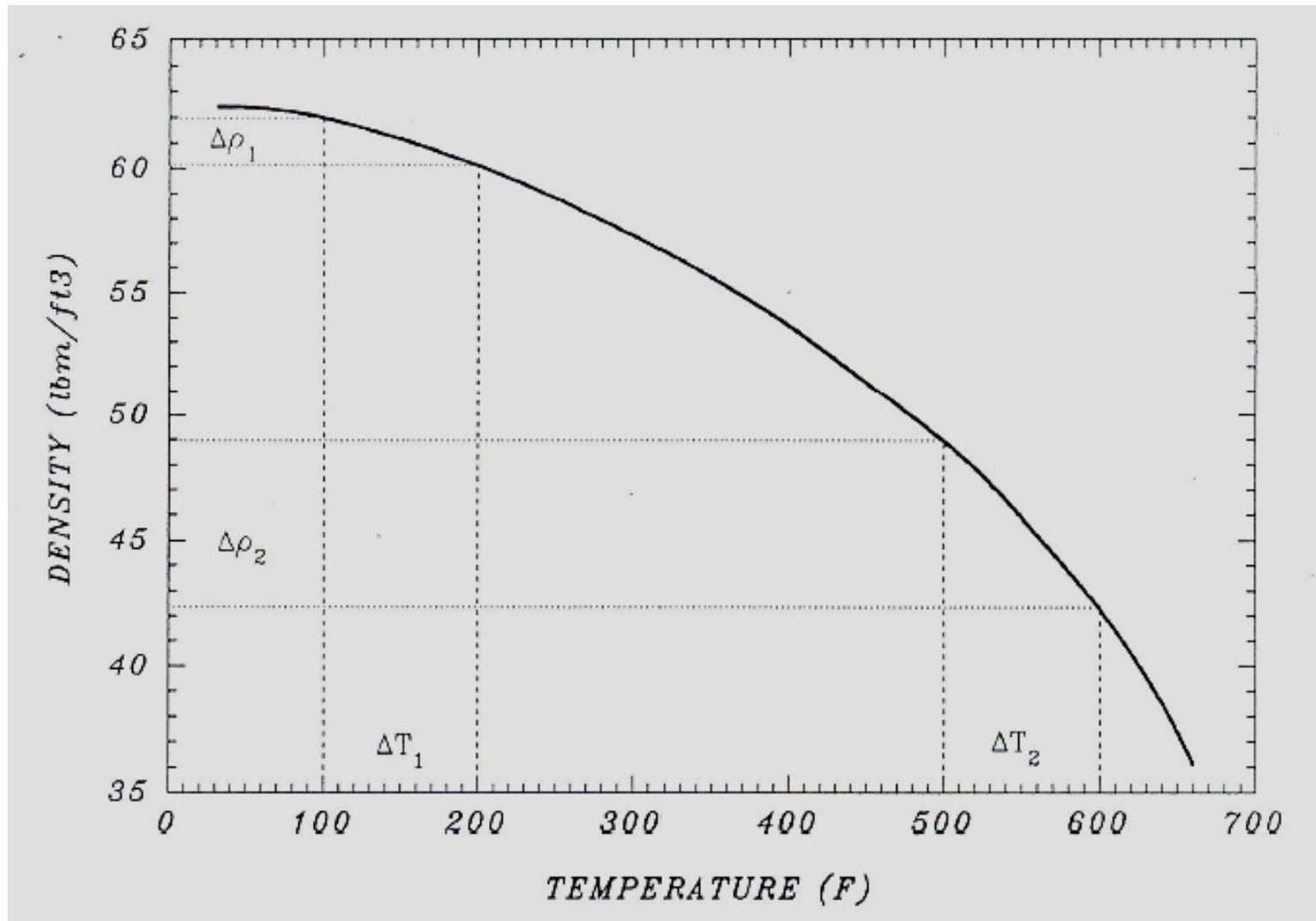
Atom density N is a function of physical density, and therefore temperature (see figure on Slide #7 for water)

Therefore if $T_{\text{fuel}} \uparrow$ (*increases*); $N_{\text{fuel}} \leftrightarrow$ (*stays the same!*) $\therefore \Sigma_{a \text{ fuel}} \leftrightarrow$

But if $T_{\text{coolant/mod}} \uparrow$; $N_{\text{coolant/mod}} \downarrow$ (*decreases*), $\therefore \Sigma_{a \text{ coolant/mod}} \downarrow$

So, f increases $\therefore \alpha_T(f) = (+) \frac{pcm}{^\circ T} = \text{Positive feedback with temp!}$

Density of Water –vs- Temperature



Moderator Temperature Coefficient

#2: Resonance Escape Probability (p):

For a liquid moderator/ coolant:

As T_{mod} $\uparrow$ (increases), N_{mod} $\downarrow$ $\therefore \Sigma_{s \text{ mod}} = N\sigma_s$ $\downarrow$ & mean thermal energy $\uparrow$

So, neutrons don't slow down as readily, spending more time in the resonance energy region.

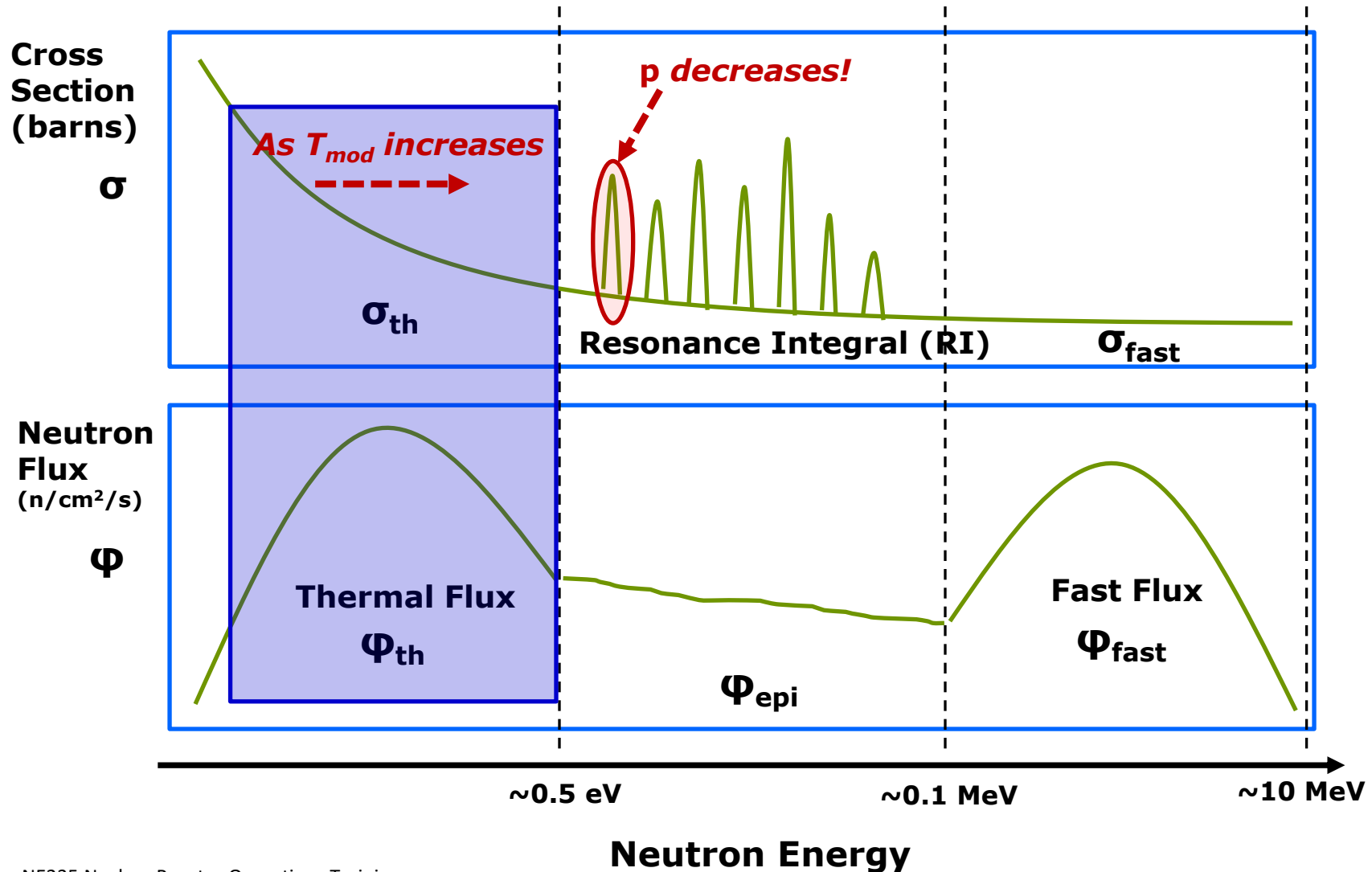
Therefore:

Capture of neutrons in resonances of ^{238}U increases (See Fig. on Slide #9)

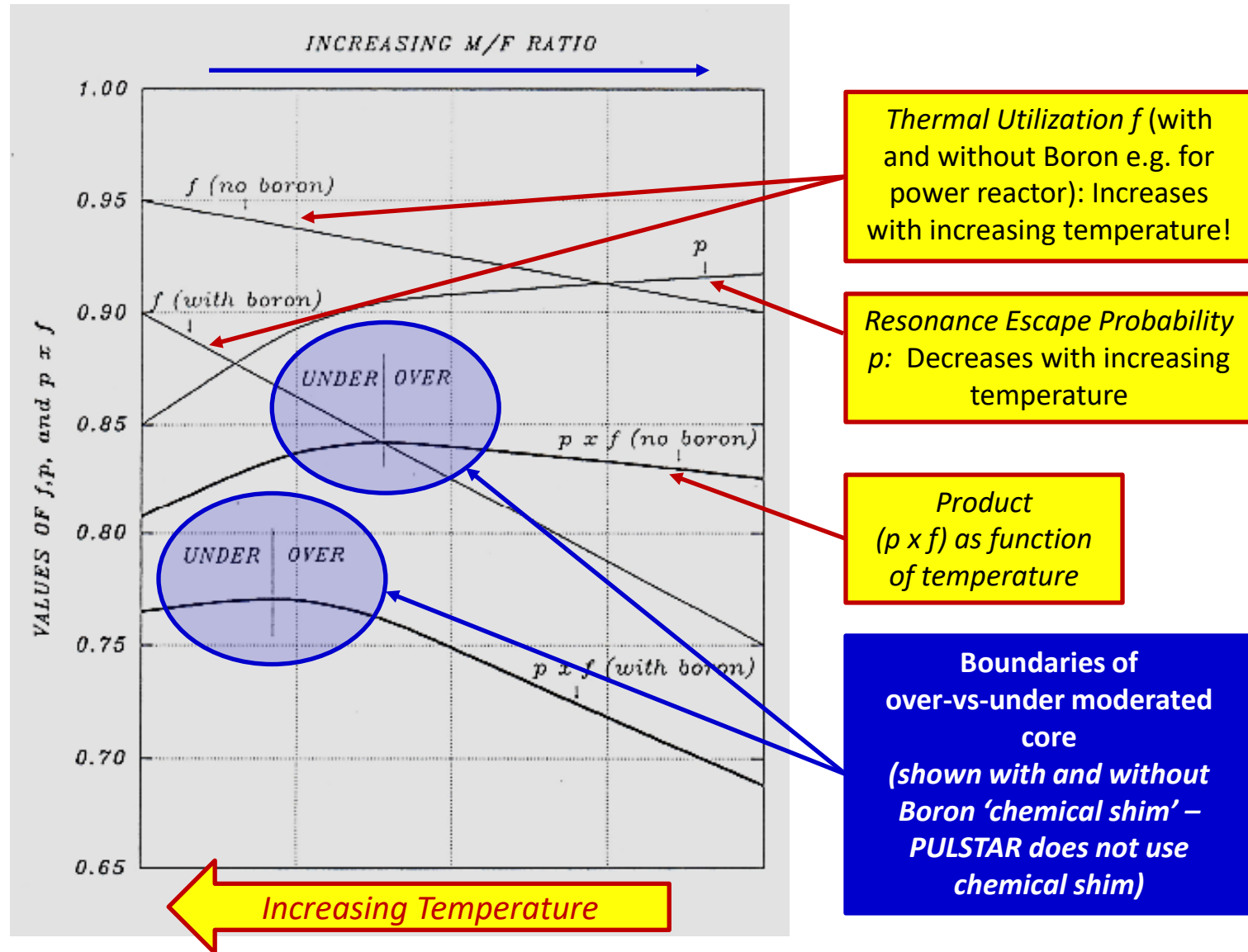
So, resonance escape probability p decreases (non linearly with temp – see Fig. on Slide 10)

$$\therefore \alpha_T(\rho) = (-) \frac{pcm}{^\circ T} = \text{negative feedback with temp!}$$

Slide #9 - Neutron Energy Spectrum and energy dependent reaction Cross Section



Slide #10 - Moderator Temperature Effects on (f) and (p)



Moderator Temperature Coefficient

#3: Thermal and Fast Non-Leakage Probability

As $T_{mod} \uparrow$ $N_{mod} \downarrow$. Therefore $\Sigma_s = N\sigma_s \downarrow$ because there are less moderator atoms to scatter neutrons. Then, leakage increases so non-leakage probability ($\mathcal{L}$) decreases.

$$\therefore \alpha_T(\mathcal{L}) = (-) \frac{pcm}{^\circ T} = \text{negative feedback with temp!}$$

So:

$$\alpha_{T mod} = \underbrace{\alpha_T(f)}_{\substack{\text{\#1: Always} \\ (+)}} + \underbrace{\alpha_T(p)}_{\substack{\text{\#2: Always } (-) \\ \text{But non-linear} \\ \text{With temp}}} + \underbrace{\alpha_T(\mathcal{L})}_{\substack{\text{\#3: Always} \\ (-)}} = \frac{\Delta\rho(pcm)}{T mod (^\circ F)}$$

So is $\alpha_{T Mod}$ Positive or Negative?

It depends on relative magnitudes of $\alpha_T(f)$ vs $\alpha_T(p)$. It's a function of fuel type, dimensions, and moderator temperature.

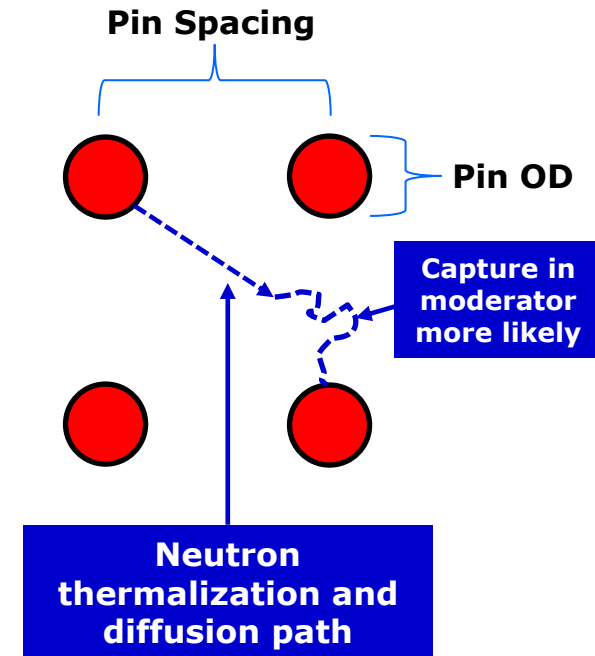
Moderator Temperature Coefficient

Case #1: Over-Moderated (High Mod/Fuel Ratio)

If pin spacing is greater than the mean thermal diffusion length then the capture in the moderator is high. So, as $T_{\text{mod}} \uparrow$ $N_{\text{mod}} \downarrow$ and $\Sigma_a \downarrow$. Therefore, the thermal utilization factor (f) increase predominates and $\alpha_{T_{\text{mod}}}$ is **positive**. This gives a **positive temperature feedback**.

So what can happen?

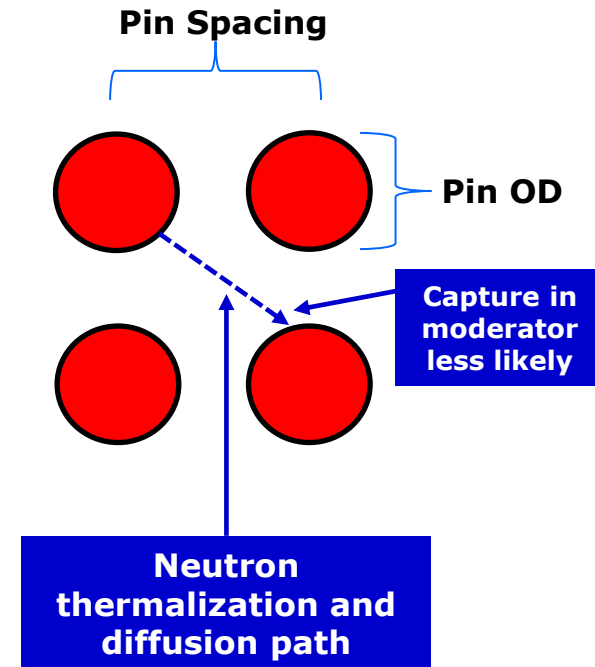
- Voids from boiling or experiments can result in positive reactivity feedback which can lead to an **uncontrolled power increase**.



Moderator Temperature Coefficient

Case #2: Under-Moderated (Low Mod/Fuel Ratio)

If the pin spacing is less than the mean thermal diffusion length, then capture in the moderator is less likely. So, as $T_{mod} \uparrow$ $N_{mod} \downarrow$ and Σ_a & $\Sigma_s \downarrow$. Therefore, the thermal utilization factor (f) does increase. However, the resonance escape probability (p) decreases, predominates, and causes $\alpha_{T_{mod}}$ to be negative. This gives a *negative temperature feedback* which serves to stabilize reactor power.



For the PULSTAR, $\alpha_{T_{mod}} = -3.9 \text{ pcm}/^{\circ}\text{F}$

Void Coefficient

B. Void Coefficient : α_{voids}

Void coefficient has the same physics as $\alpha_{T \text{ mod}}$. Displacement of the moderator, whether by thermal expansion or by the introduction of voids gives the same effect on reactivity. Therefore, we want a negative void coefficient.

Where might voids come from in a power reactor?

Scenario: A PWR power reactor is operating at full power and temperature when a loss of primary pressurization occurs. Therefore, primary coolant flashes to steam, which leads to boiling which then leads to voids.

Void Coefficient

Consider the scenario:

If the PWR is under-moderated, voids will result in **negative reactivity leading to lower power**. If it is over-moderated, there will be a **positive reactivity effect leading to higher power**.

For the PULSTAR, $\alpha_{voids} = -1.6 \text{ pcm/cc}_{void}$

The PULSTAR is not pressurized, so is not susceptible to a loss of pressurization accident. However, the primary water could boil if coolant channel was blocked or if a pressurized experiment ruptures and releases gas bubbles.

So, we design reactors to be under-moderated under all credible operating and accident scenarios.

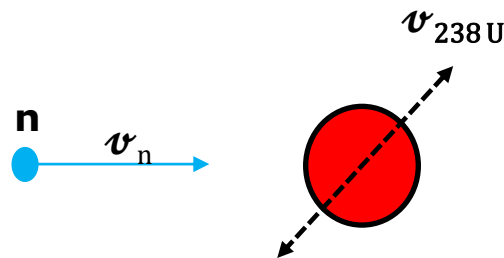
Fuel Temperature Coefficient (Doppler)

C. Fuel Temperature Coefficient (Doppler Coefficient) : $\alpha_{T \text{ fuel}}$

$$\alpha_{T \text{ fuel}} = \frac{\Delta pcm}{\Delta T_{\text{fuel}}}$$

Doppler effect $\rightarrow$ concept of relative motion of neutrons -vs- ^{238}U

In “rest” frame:



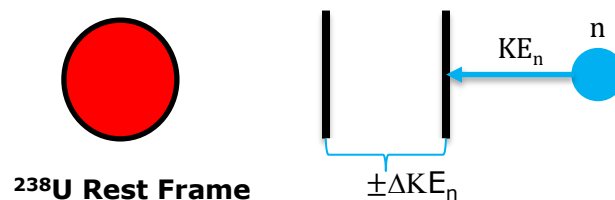
Thermal motion of ^{238}U atom is a function of fuel temperature where

$$v \propto \sqrt{T}$$

where $KE = \frac{1}{2} mv^2$

Fuel Temperature Coefficient (Doppler)

In the frame of the absorbing ^{238}U nucleus:

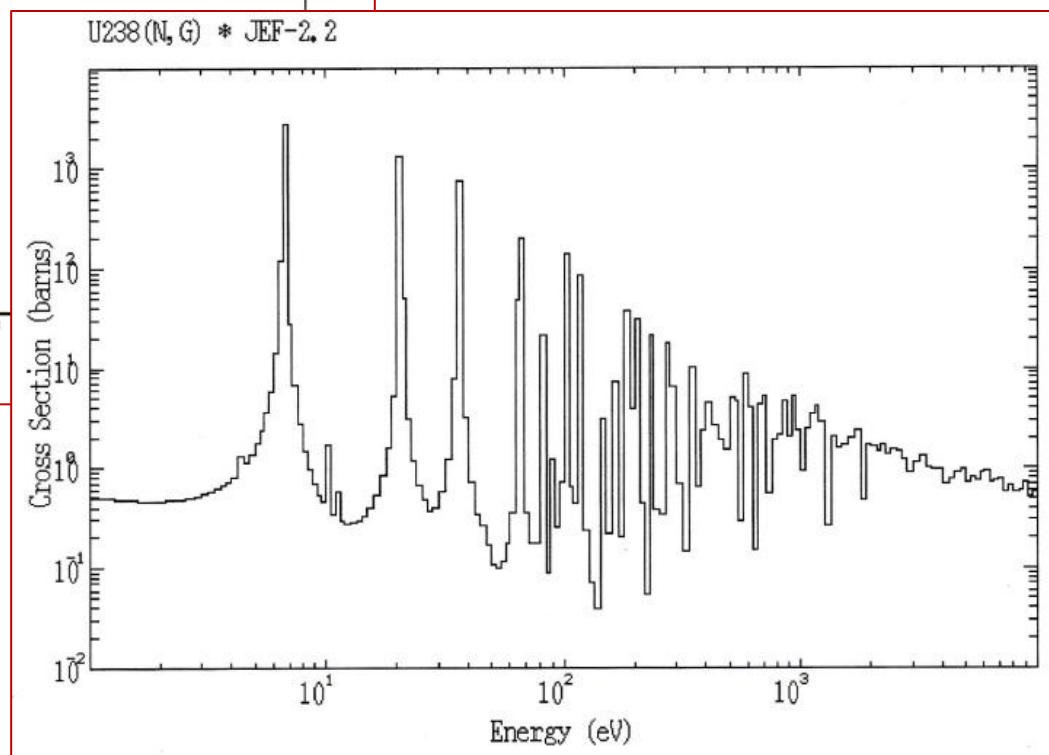
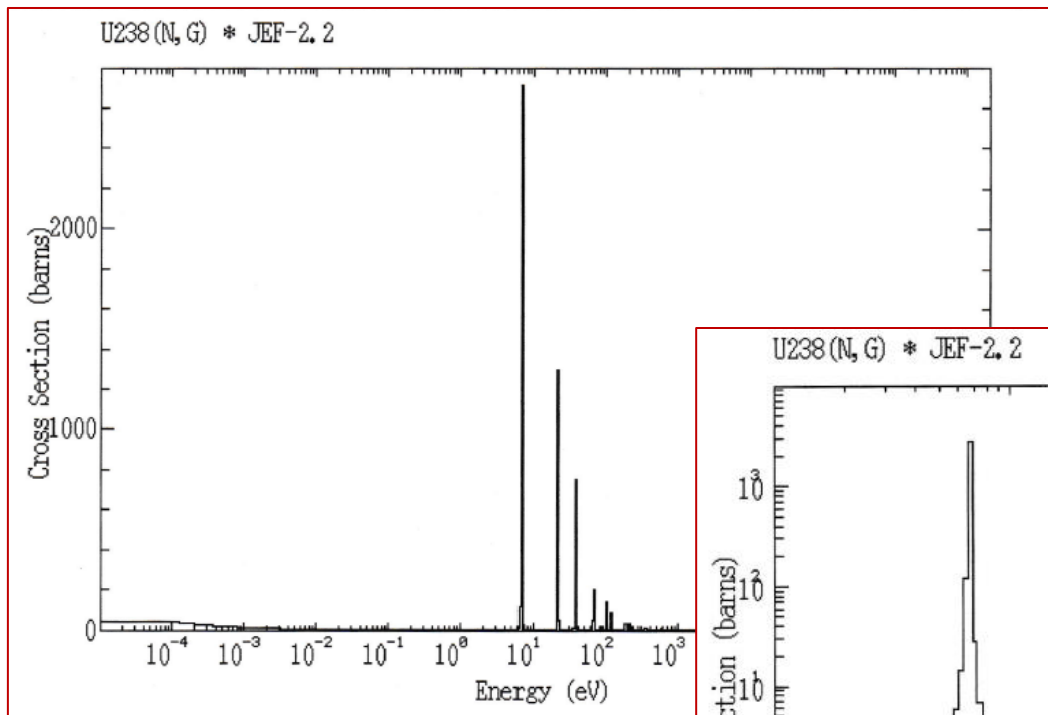


Kinetic energy of the neutron is $KE_n \pm \Delta KE_n$ due to relative motion.

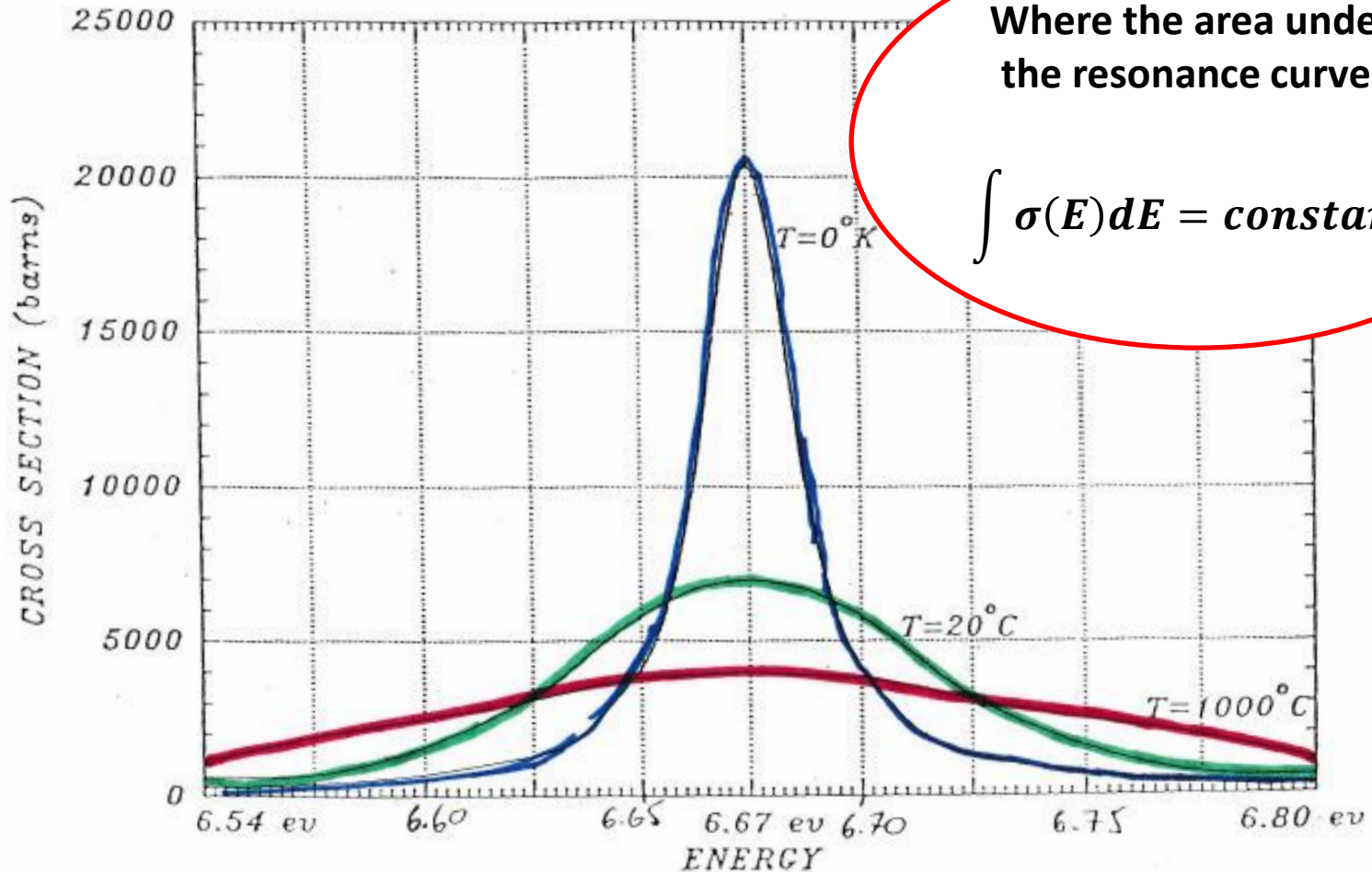
^{238}U cross section $\Rightarrow$ function of neutron KE (See Fig. on Slide #18)

Therefore, the shape of resonances “broadens” with increased temperature, while total area under the resonance stays constant (See Fig. on Slide #19).

^{238}U Absorption Cross Section

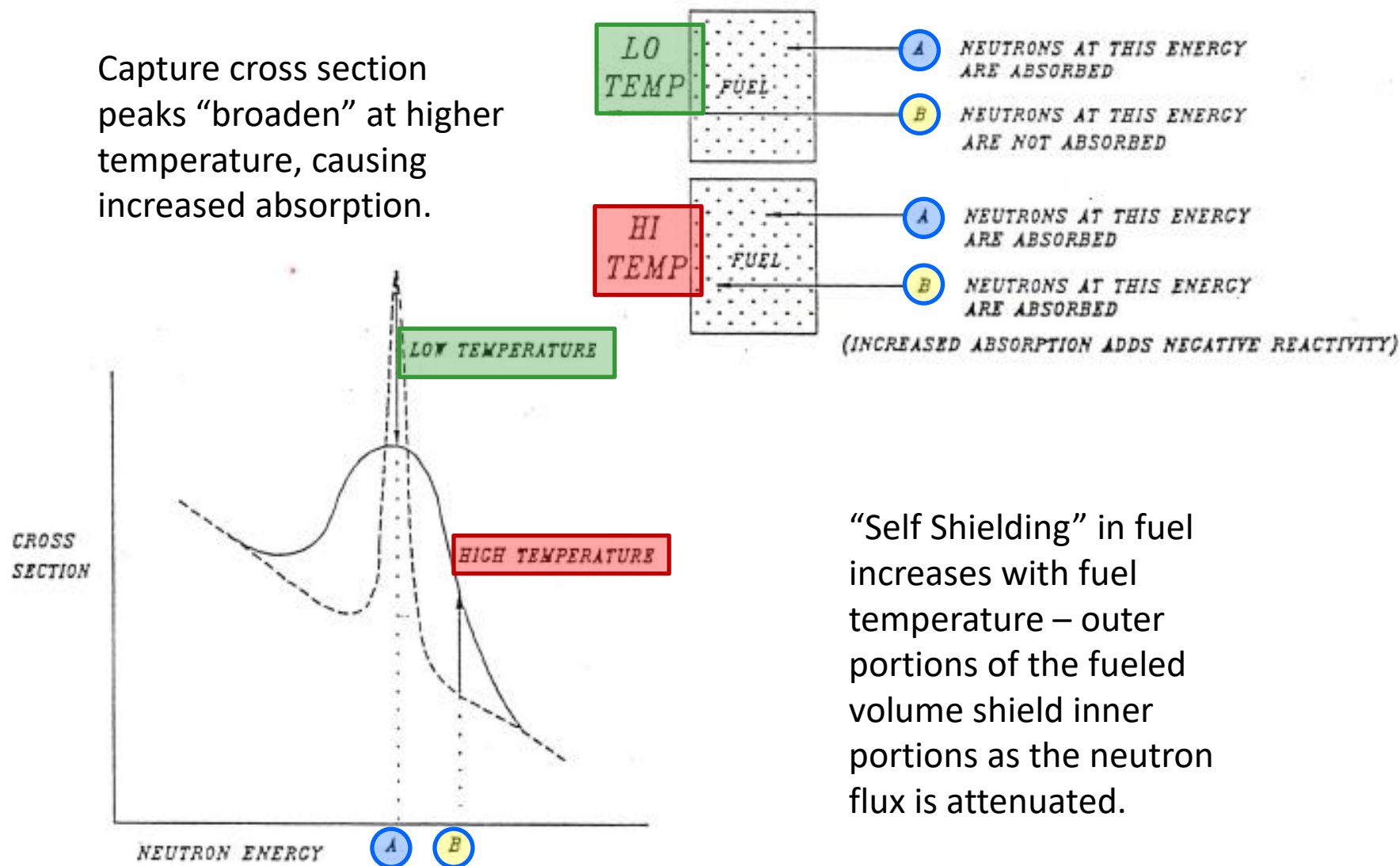


Doppler Broadening



Doppler Broadening & Self Shielding

Capture cross section peaks “broaden” at higher temperature, causing increased absorption.



“Self Shielding” in fuel increases with fuel temperature – outer portions of the fueled volume shield inner portions as the neutron flux is attenuated.

Fission Product Poisons

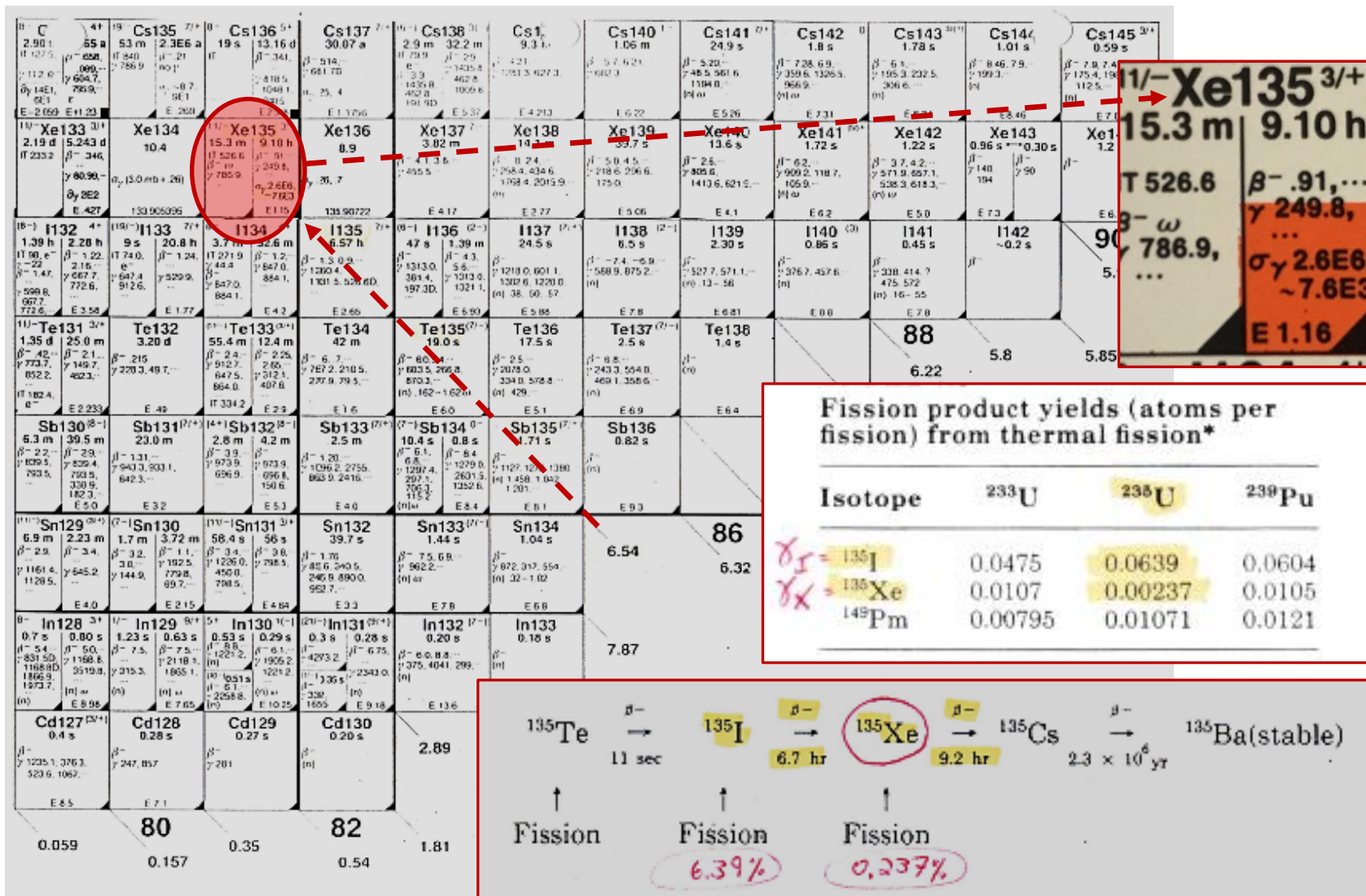
- D. There are two primary isotopes that are fission product poisons, ^{135}Xe & ^{149}Sm , both with a large absorption cross section (σ_a).

^{135}Xe (Xenon-135):

$^{135}\text{Xe} \sigma_a = 2.6 \times 10^6 \text{ barns}$ -vs- $^{235}\text{U} \sigma_f = 585 \text{ barns}$.
4400x greater!!

See Slide #22 for Production Modes

Slide #22 – ^{135}Xe Production



^{135}Xe Production

^{135}Xe activity as a function of time and power:

$$^{135}\text{Xe} \propto \lambda_I ^{135}\text{I} + \gamma_x \bar{\Sigma}_f \bar{\Phi}_T - \lambda_{\text{Xe}} \text{Xe} - \sigma_{a_x} \bar{\Phi}_T ^{135}\text{Xe}$$

Sources		Sinks	
Iodine Decay to Xenon (97%)	Direct Fission Production of Xenon (~3%)	Decay of Xenon	Burn-up of Xenon

^{135}I activity as a function of time and power:

$$^{135}\text{I} \propto \lambda_{\text{Te}} \text{Te} + \gamma_I \bar{\Sigma}_f \bar{\Phi}_T - \lambda_I ^{135}\text{I}$$

Tellurium Decay to Iodine	Fission Production of Iodine	Decay of Iodine
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^{135}Xe Production

Where:

X_e = Xe-135 activity

I = Iodine-135 activity

σ_{a_x} = 2.6×10^6 barns

γ_x = Fission yield of Xenon (0.24% for U-235)

γ_I = Fission yield of Iodine (6.39% for U-235)

$\bar{\Sigma}_f$ = Average thermal macroscopic fission cross section

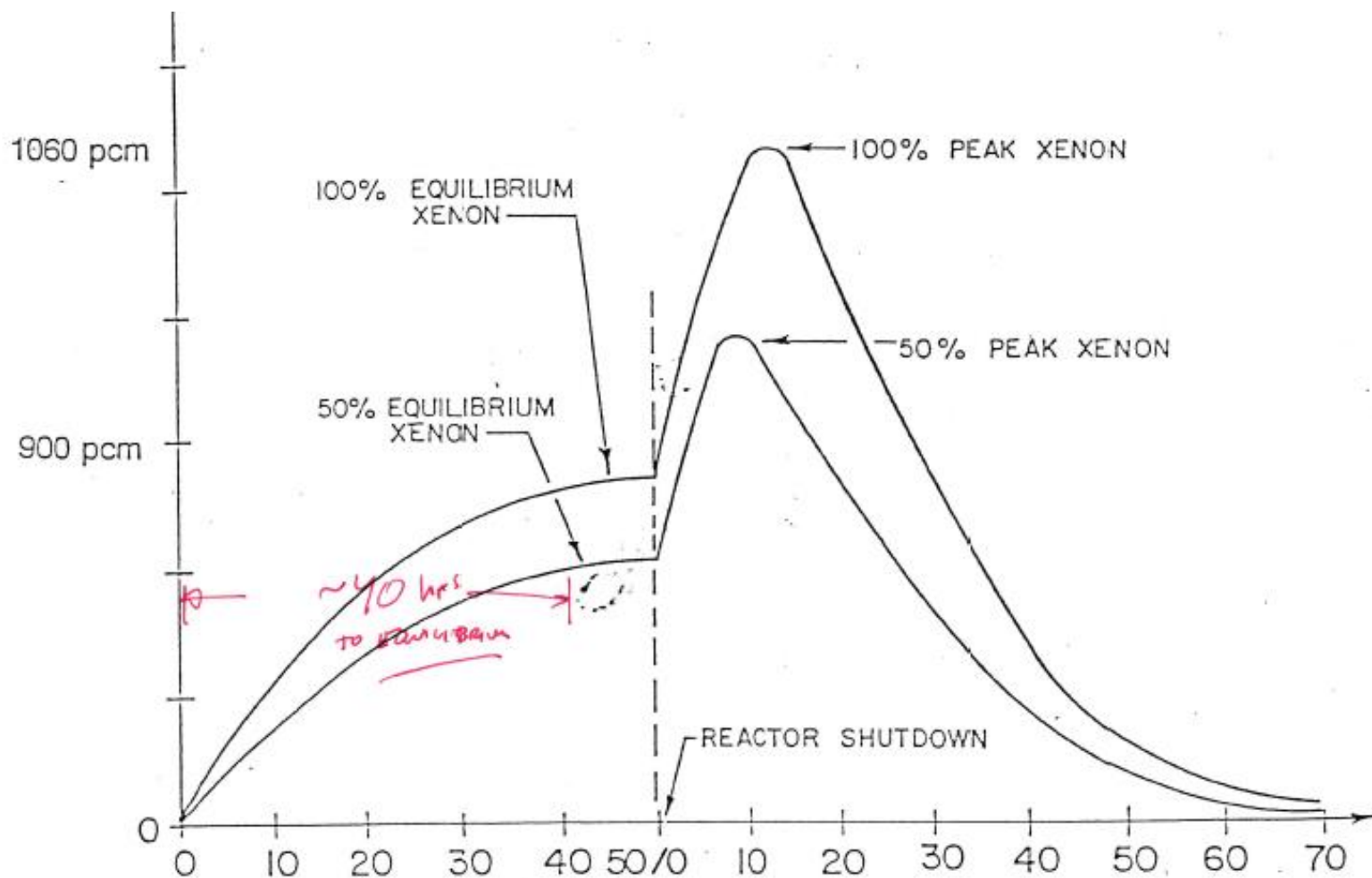
$\bar{\Phi}_T$ = Average thermal flux

λ_I & λ_X = Decay constants

X_e , I , $\bar{\Sigma}_f$, and $\bar{\Phi}_T$ change throughout the core matrix due to gradients in core loading, flux and power density.

For PULSTAR, equilibrium $\rho_{Xe} \sim -900$ pcm.

^{135}Xe Production – Time & Power Dependent



^{135}Xe Production – Time & Power Dependent

Xenon Concentration:

- A.** Equilibrium concentration or activity is reached after ~5 half-lives (see Figure on Slide #25):

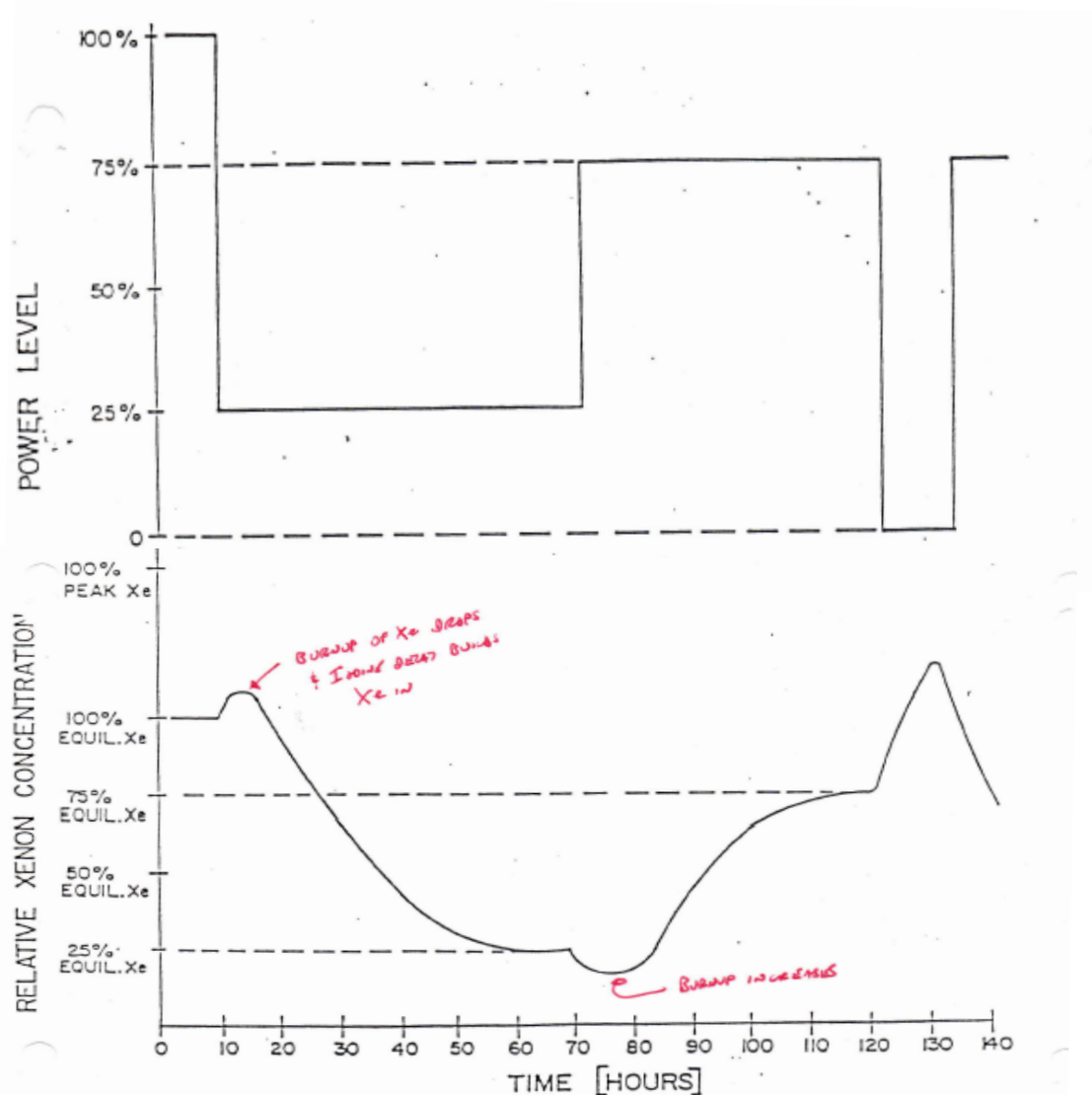
$$t_{1/2} \text{ } ^{135}\text{Xe} = 9.1 \text{ hours} \times 5 \simeq 40 - 45 \text{ hours to equilibrium}$$

- B.** Equilibrium concentration is a function of flux & power. Therefore, it saturates at a higher level for higher power.

- C.** If power is decreased (see Figure on Slide #27):

- 1.** Burn-up decreases while production from decay of iodine is unchanged. Therefore, Xenon increases.
- 2.** Iodine decays towards new lower equilibrium concentration. Therefore, Xenon decreases and will peak 2-6 hours after power is decreased or reactor is shutdown.

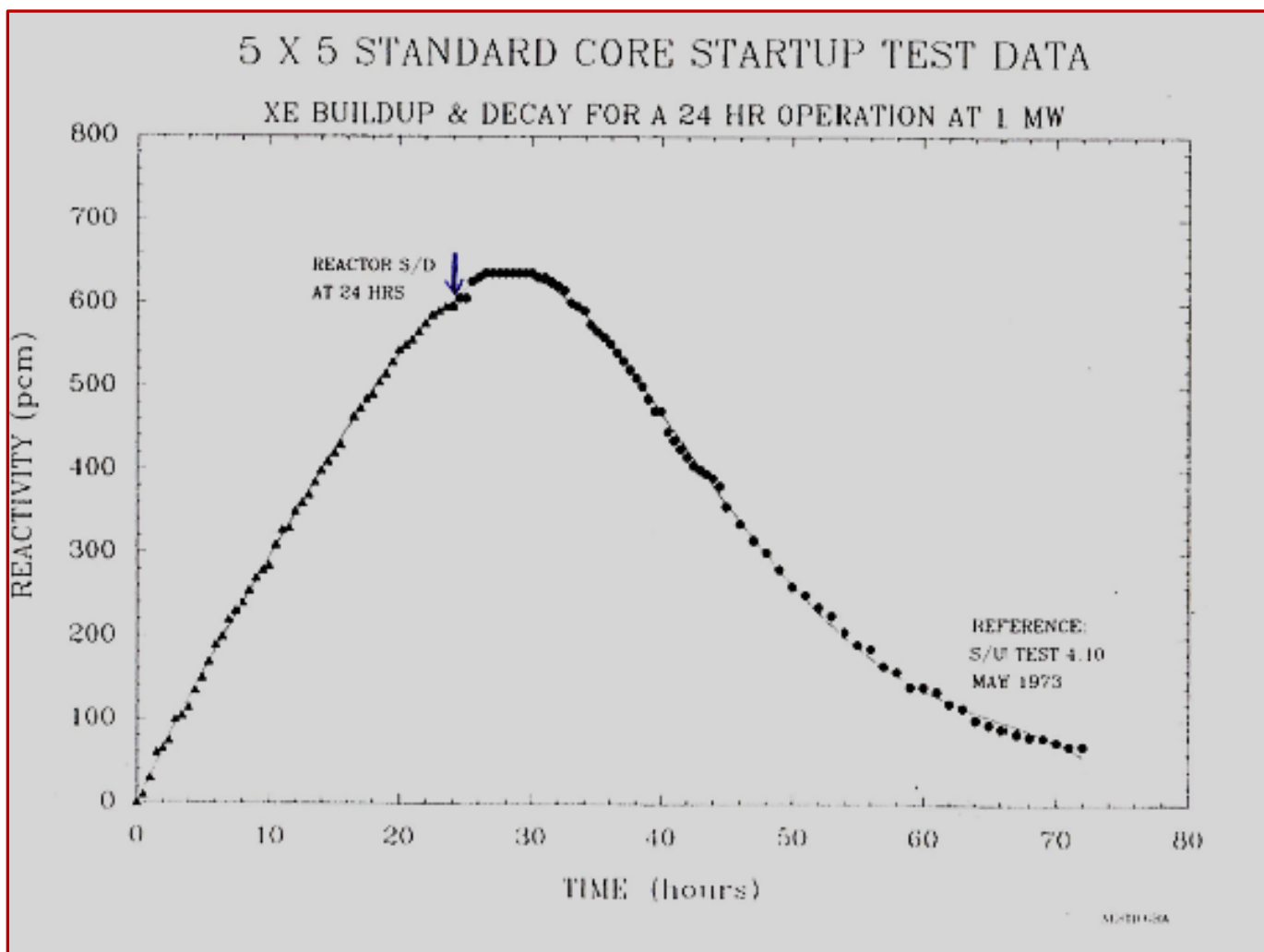
^{135}Xe Production – Time & Power Dependent



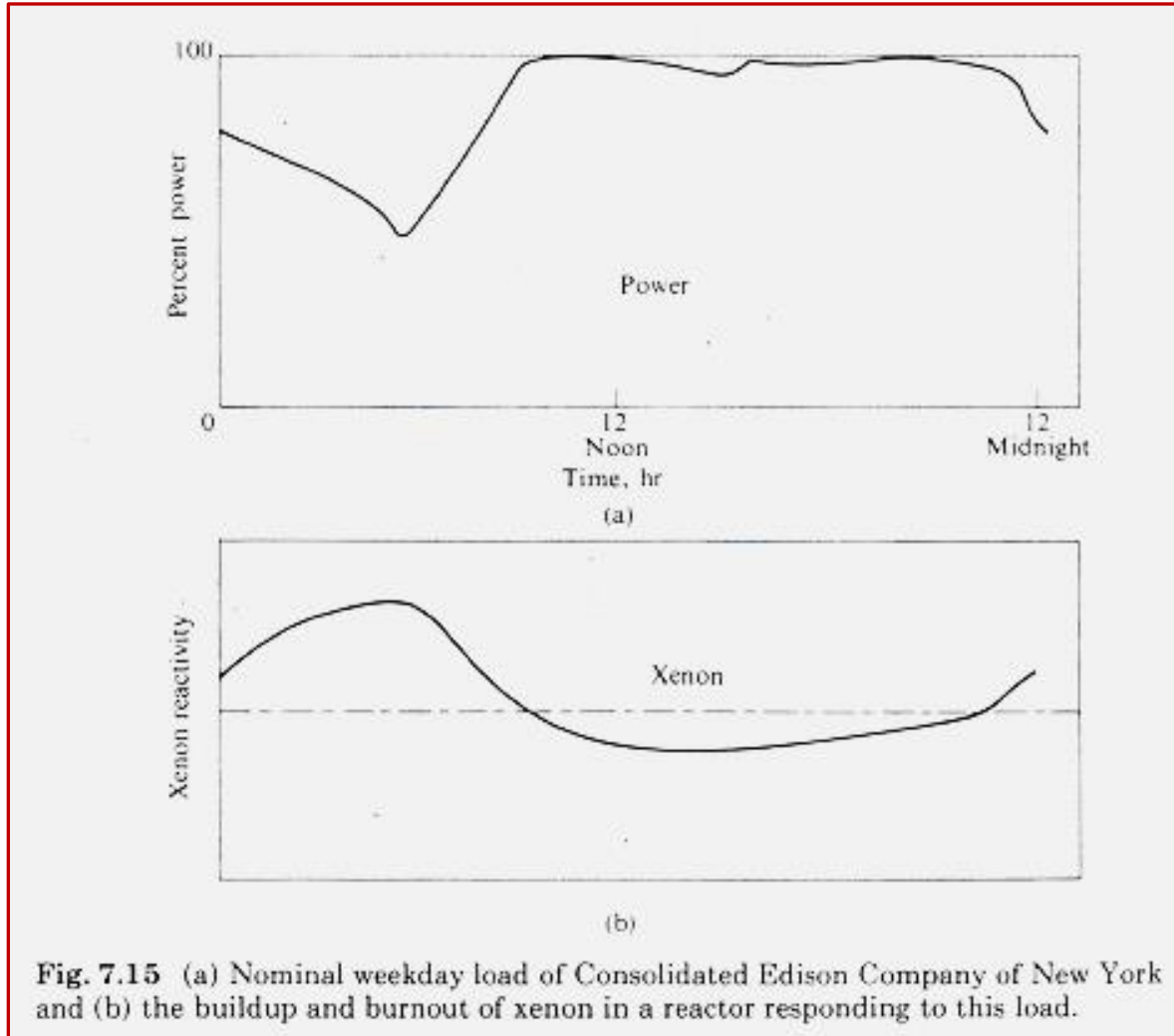
^{135}Xe Production – Time & Power Dependent

- D.** If power is increased (see Figure on Slide #27):
- 1.** Burn-up will increase while ^{135}I concentration is initially unchanged. Therefore, Xenon decreases and reaches a minimum 2-6 hours after the start of the increase.
 - 2.** ^{135}I concentration builds up which counteracts burn-up. Therefore, Xenon increases until reaching a new equilibrium value in 40-45 hours.
- E.** If reactor is shutdown Xenon peaks within 2-6 hours (depending on operating history) then decays to Xenon free in ~70 hours.

^{135}Xe Production – PULSTAR Startup Testing



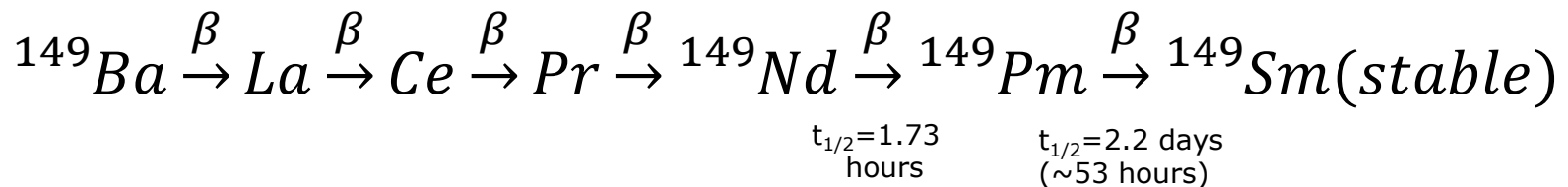
Xenon in Load Following Power Plant



Fission Product Poisons

Fission Product Poisons: ^{149}Sm

$$^{149}\text{Sm} \sigma_a = \mathbf{4 \times 10^4 \text{ barns}}; \gamma_{\text{Sm}} \sim 1.1\%$$



$$\therefore \text{Sm} \propto \lambda_{\text{Pm}} \text{Pm} + \gamma_{\text{Sm}} \bar{\Sigma}_f \bar{\Phi}_T - \sigma_{a_{\text{Sm}}} \Phi_T \text{Sm}$$

Decay of Fission Production Burn-up of ^{149}Sm ,
 Promethium of ^{149}Sm (stable so no decay
 term)

^{149}Sm is stable and is only removed by burnout.

Buildup of Poisons in a Power Reactor

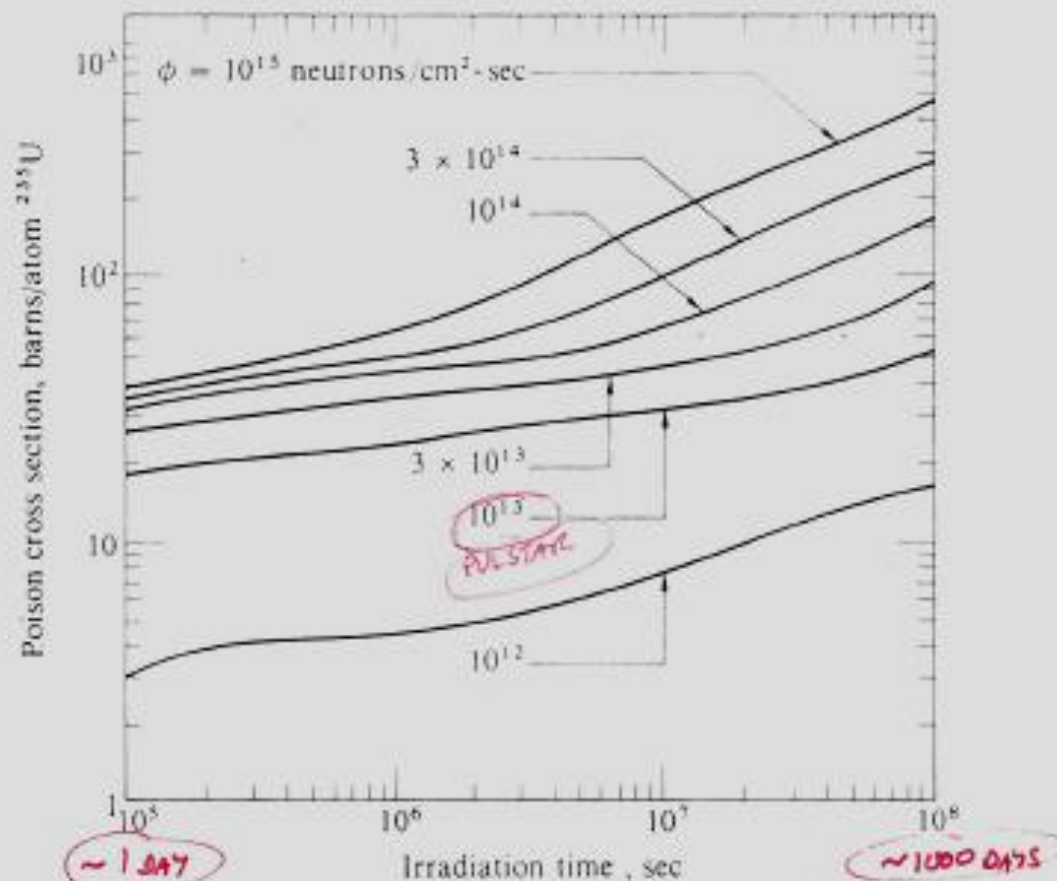


Fig. 7.17 Buildup of fission product poisons in barns of absorption cross section per original atom of ^{235}U . (From ANL-5800, 2nd ed., 1963.)

Summary

1. Moderator Temperature Coefficient – design reactor so that $\alpha_{T_{mod}}$ is negative for all operating and accident conditions.
2. Void Coefficient – same physics as $\alpha_{T_{mod}}$; assure negative feedback.
3. Doppler Broadening (Fuel Temperature) Coefficient – provides significant negative feedback in low enriched reactors (e.g. with higher ^{238}U content in fuel).
4. Fission Product Poisons – Xenon-135 contributes significant negative reactivity as a function of power history.